

LipidIN × Agent — `code_for_PB` User Manual (English)

Paternò-Büchi (PB) reaction pipeline for **carbon-carbon double-bond position localization** in lipids, driven either as a one-shot batch script or through a conversational web agent.

A Chinese version of this manual is in [README_ZH.md](#).

This manual is written from an actual read of the whole pipeline and a real run of the bundled demo. All screenshots below are from the live interface; the figures are produced by the demo data.

1. What this does

Conventional lipidomics tells you *how much* of a lipid species (e.g. `PC 42:8`) is present but not *where* its double bonds sit. PB derivatization tags each C=C double bond so it can be pinpointed. This pipeline:

- Stage 1 — Conventional annotation** of the underivatized sample (`common/` , negative mode, acetate library) → identifies species and quantifies them.
- Stage 2 — PB search** of the derivatized sample (`PB/` , positive `[M+H]+` , PB library) → finds the diagnostic ion pairs.
- Stage 3 — Double-bond localization** → reconstructs positions in omega notation (`n-9` , `n-6` , ...).
- Stage 4 — Chain × double-bond-position quantification** with total-area normalization.

The web agent adds downstream steps: experiment design, QC + differential analysis (species layer **and** double-bond-position layer), publication figures, LLM interpretation, hypotheses (literature + LIPID MAPS), verification, and a Markdown report — **11 stages** in total.

2. Requirements

Item	Requirement
OS / Python	Windows x64 + CPython 3.11 (the C++ engines ship as <code>*.cp311-win_amd64.pyd</code>)
RAM	≥ 16 GB (the theoretical libraries are large)
Python packages	<code>flask pandas numpy scipy scikit-learn matplotlib polars plotly pybind11 fisher_py</code>
Thermo <code>.raw</code> reading	<code>fisher_py</code> + .NET runtime (bundled with fisher_py)
Spectral libraries	Download from Zenodo → https://zenodo.org/records/21421866 (<code>PB_PX_H.pkl</code> , <code>common_PX_CH3C00.pkl</code> , <code>common_PX_CH3C00.pkl.ms1index.pkl</code>)
LLM (optional)	An API key for Anthropic / OpenAI-compatible / DeepSeek. Without it, structured commands still work; only free-form design/interpretation/hypotheses need it.

Put the three library files into a folder (e.g. `PB library/`). Note: `fisher_py` reads Thermo `.raw` directly — only **strict DDA** acquisitions are accepted (DIA/SWATH/PASEF are rejected on purpose).

3. Prepare your data

The pipeline needs **two acquisitions of the same samples**, in two subfolders:

```
<raw root>/
├ common/      # underivatized, negative mode → MSMS1.raw, MSMS2.raw, ...
└ PB/          # PB-derivatized, positive mode → PB1_A1.raw, PB2_A1.raw, ...
```

- `PB1_*` and `PB2_*` are **two technical injections of the same sample** — the pipeline merges them automatically in Stage 4.
- The bundled demo lives at `D:\bio_inf\LipidIN_github\PB_demo\demo` (with `common/` and `PB/`).

4. Two ways to run

Option A — One-shot batch (`run.py`)

Open `run.py` and edit the path block near the top (`PROJECT_ROOT` / `SAMPLE_ROOT` / `LIBRARY_DIR` , or point `RAW_ROOT` at a folder containing `common/` and `PB/`). Then:

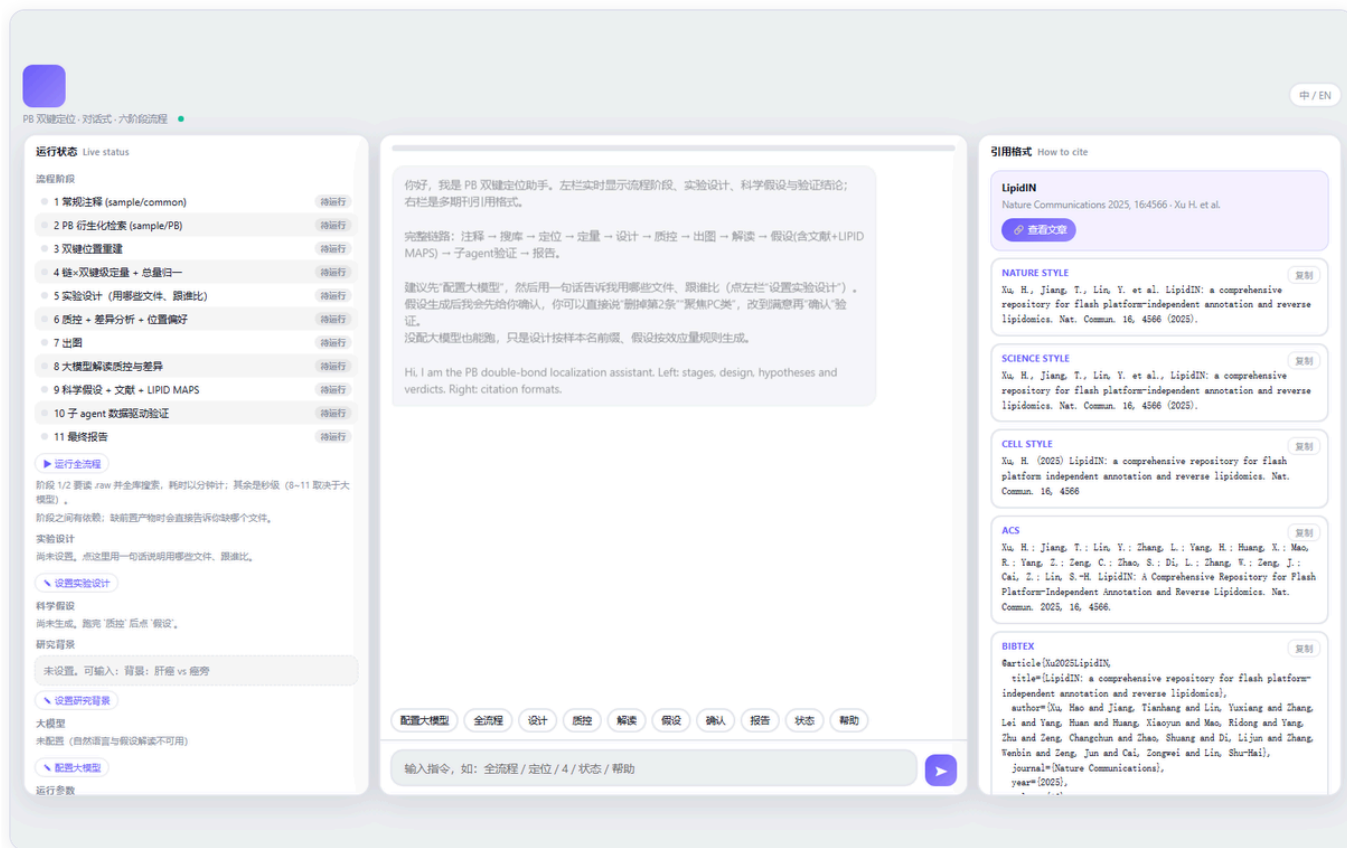
```
D:\anaconda3\python.exe run.py
```

It runs Stage 1 → 4 in order and writes all CSV outputs into each `.../pk1/` folder. Toggle `RUN_STAGE1..4` at the top to run only part of the pipeline. Keep `EQ3_WORKERS = 1` on a 16 GB machine.

Option B — Conversational agent (recommended)

```
D:\anaconda3\python.exe pb_agent_server.py --no-open --port 8790
# then open http://127.0.0.1:8790 in your browser
```

You get a three-pane interface: **left** = live stage status + parameters, **center** = chat, **right** = citation formats.



5. A full standard conversation

Below is a real end-to-end session against the demo. Type these into the chat box (or click the chips).

5.1 Ask for help — 帮助 / help

The agent lists every command and the full path `annotation → search → localize → quantify → design → qc → plots → interpret → hypothesis → verify → report` (you can also type the stage number `1 – 11`).

运行状态Live status

流程阶段

1 常规注释 (sample/common)

2 PB 衍生化检查 (sample/PB)

3 双键位置重建

4 链+双键定量 + 总量归一

5 实验设计 (用哪些文件、跟谁比)

6 质控 + 差异分析 + 位置偏好

7 出图

8 大模型解读质控与差异

9 科学假设 + 文献 + LIPID MAPS

10 子 agent 数据驱动验证

11 最终报告

运行全流程

阶段 1/2 要读 raw 并全屏渲染, 耗时以分钟计; 其余是秒级 (8~11 取决于大模型)。

阶段之间有依赖: 缺前置产物时会直接告诉你缺哪个文件。

实验设计

尚未设置, 点这里用一句话说明用哪些文件、跟谁比。

设置实验设计

科学假设

尚未生成, 跑完“质控”后点“假设”。

研究背景

未设置, 可输入: 背景: 肝癌 vs 癌旁

设置研究背景

大模型

未配置 (自然语言与假设解读不可用)

配置大模型

你好, 我是 PB 双键定位助手。左栏实时显示流程阶段、实验设计、科学假设与验证结论; 右栏是多期刊引用格式。

完整链路: 注释 → 搜库 → 定位 → 定量 → 质控 → 出图 → 解读 → 假设(含文献 + LIPID MAPS) → 子 agent 验证 → 报告。

建议先“配置大模型”, 然后用一句话告诉我用哪些文件、跟谁比 (点左栏“设置实验设计”)。

假设生成后我会先给你确认, 你可以直接说“删掉第2条”“聚焦PC类”, 改到满意再“确认”验证。

没配大模型也能跑, 只是设计按样本名前缀、假设按效应量规则生成。

Hi, I am the PB double-bond localization assistant. Left: stages, design, hypotheses and verdicts. Right: citation formats.

我可以把 PB 双键定位从原始数据一路跑到带验证的报告。

先告诉我原始文件在哪: “raw 根目录 D:\...” (其下要有 common/ 与 PB/ 两个子目录), raw → pkl 由阶段 1/2 自动完成

分步: “annotation” / “search” / “localize” / “quantify” / “design” / “qc” / “plots” / “interpret” / “hypothesis” / “verify” / “report” (也可输入序号 1~11)

“全流程”依次跑完 (跑到“假设”会停下来等你确认)

“配置大模型”—— 配好才能用自然语言说设计、才有真正的解读与假设

“设计”—— 说明用哪些文件、跟谁比, 例如“只用 A1 到 A3 和 B1 到 B3, A 是病例”

“背景: 肝癌 vs 癌旁”—— 告诉我这是什么样本, 假设会更贴题

假设列出来后: “确认”开始验证, 或直接说怎么改 (“删掉第2条”“聚焦 PC 类”)

“状态”看进度

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输入指令, 如: 全流程 / 定位 / 4 / 状态 / 帮助

引用格式How to cite

LipidIN

Nature Communications 2025, 16:4566 · Xu H. et al.

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5.2 Point it at your raw data — raw root D:\ ...

raw root D:\bio_inf\LipidIN_github\PB_demo\demo

The agent detects common/ and PB/, updates all derived paths, and the left panel refreshes. Stages whose products already exist on disk are marked 盘已有 (on disk).

运行状态Live status

流程阶段

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6 质控 + 差异分析 + 位置偏好

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8 大模型解读质控与差异

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实验设计

尚未设置, 点这里用一句话说明用哪些文件、跟谁比。

设置实验设计

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尚未生成, 跑完“质控”后点“假设”。

研究背景

未设置, 可输入: 背景: 肝癌 vs 癌旁

设置研究背景

大模型

未配置 (自然语言与假设解读不可用)

配置大模型

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“设计”—— 说明用哪些文件、跟谁比, 例如“只用 A1 到 A3 和 B1 到 B3, A 是病例”

“背景: 肝癌 vs 癌旁”—— 告诉我这是什么样本, 假设会更贴题

假设列出来后: “确认”开始验证, 或直接说怎么改 (“删掉第2条”“聚焦 PC 类”)

“状态”看进度

配置大模型 全流程 设计 质控 解读 假设 确认 报告 状态 帮助

输入指令, 如: 全流程 / 定位 / 4 / 状态 / 帮助

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```
[config] run.py 已更新: ['RAW_ROOT': 'D:\\bio_inf\\LipidIN_github\\PB_demo\\demo', 'COMMON_RAW_DIR': 'D:\\bio_inf\\LipidIN_github\\PB_demo\\demo\\common', 'COMMON_OUT_DIR': 'D:\\bio_inf\\LipidIN_github\\PB_demo\\demo\\common\\pkl', 'PB_RAW_DIR': 'D:\\bio_inf\\LipidIN_github\\PB_demo\\demo\\PB', 'PB_OUT_DIR': 'D:\\bio_inf\\LipidIN_github\\PB_demo\\demo\\PB\\pkl']
```

注意: 这些阶段的产物是用旧参数跑的, 要重跑才作数 —— 2 PB 衍生化检查 (sample/PB)、3 双键位置重建、4 链+双键定量 + 总量归一

5.3 Check progress anytime — 状态 / status

运行状态 Live status

流程阶段

1 常规注释 (sample/common)

2 P8 衍生化检索 (sample/P8)

3 双键位置重建

4 链+双键级定量 + 总量归一

5 实验设计 (用哪些文件、跟谁比)

6 质控 + 差异分析 + 位置偏好

7 出图

8 大模型解读质控与差异

9 科学假设 + 文献 + LIPID MAPS

10 子 agent 数据驱动验证

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▶ 运行全流程

阶段 1/2 阅读 raw 并全库搜索, 耗时以分钟计; 其余是秒级 (8~11 取决于大模型)。

阶段之间有效数: 缺前置产物时会直接告诉你缺哪个文件。

实验设计

尚未设置, 点这里用一句话说明用哪些文件、跟谁比。

设置实验设计

科学假设

尚未生成, 跑完“质控”后点“假设”。

研究背景

未设置, 可输入: 背景: 肝癌 vs 癌旁

设置研究背景

大模型

未配置 (自然语言与假设解读不可用)

配置大模型

• 假设列出来后: “确认”开始验证, 或直说怎么改 (| 删掉第2条 | 聚焦 PC 类)

• 状态 看进度

raw 根目录 D:\bio_inf\LipidIN_github\PB_demo\demo

已更新:

```
raw_root = D:\bio_inf\LipidIN_github\PB_demo\demo
```

```
[config] run.py 已更新: 'RAW_ROOT': 'D:\bio_inf\LipidIN_github\PB_demo\demo', 'COMMON_RAW_DIR': 'D:\bio_inf\LipidIN_github\PB_demo\demo\common', 'COMMON_OUT_DIR': 'D:\bio_inf\LipidIN_github\PB_demo\demo\common\pk1', 'PB_RAW_DIR': 'D:\bio_inf\LipidIN_github\PB_demo\demo\PB', 'PB_OUT_DIR': 'D:\bio_inf\LipidIN_github\PB_demo\demo\PB\pk1'
```

注意: 这些阶段的产物是用旧参数跑的, 要重跑才作数 —— 2 P8 衍生化检索 (sample/P8)、3 双键位置重建、4 链+双键级定量 + 总量归一

请库目录 D:\bio_inf\LipidIN_github\PB_demo\PB library

这些没改:

```
library_dir: 目录不存在: D:\bio_inf\LipidIN_github\PB_demo\PB
```

已完成: 无

待运行: 1 常规注释 (sample/common), 2 P8 衍生化检索 (sample/P8), 3 双键位置重建, 4 链+双键级定量 + 总量归一, 5 实验设计 (用哪些文件、跟谁比), 6 质控 + 差异分析 + 位置偏好, 7 出图, 8 大模型解读质控与差异, 9 科学假设 + 文献 + LIPID MAPS, 10 子 agent 数据驱动验证, 11 最终报告

配置大模型 全流程 设计 质控 解读 假设 确认 报告 状态 帮助

输入指令, 如: 全流程 / 定位 / 4 / 状态 / 帮助

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5.4 Run the whole thing — 全流程 / run all

全流程 runs Stage 1 → the hypothesis step and then **pauses for your confirmation** before verification. Or run any single stage by name/number. Here we run **plots** (出图); the agent streams the log live and reports where the figures went:

运行状态 Live status

流程阶段

1 常规注释 (sample/common)

2 P8 衍生化检索 (sample/P8)

3 双键位置重建

4 链+双键级定量 + 总量归一

5 实验设计 (用哪些文件、跟谁比)

6 质控 + 差异分析 + 位置偏好

7 出图

8 大模型解读质控与差异

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阶段 1/2 阅读 raw 并全库搜索, 耗时以分钟计; 其余是秒级 (8~11 取决于大模型)。

阶段之间有效数: 缺前置产物时会直接告诉你缺哪个文件。

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研究背景

未设置, 可输入: 背景: 肝癌 vs 癌旁

设置研究背景

大模型

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完整链路: 注释 → 检索 → 定位 → 定量 → 设计 → 质控 → 出图 → 解读 → 假设(含文献 + LIPID MAPS) → 子 agent 验证 → 报告。

建议先“配置大模型”, 然后用一句话告诉我用哪些文件、跟谁比 (点左栏“设置实验设计”)。

假设生成后我会先给你确认, 你可以直接说“删掉第2条”, 改到满意再“确认”验证。

没配大模型也能跑, 只是设计按样本名称喂, 假设按效应量规则生成。

Hi, I am the P8 double-bond localization assistant. Left: stages, design, hypotheses and verdicts. Right: citation formats.

▶ 开始 7 出图

```
[PB-fg] 已导出 D:\bio_inf\LipidIN_github\PB_demo\demo\PB\pk1\figures\Pis1_position_dimension.svg / .pdf / .png
```

```
[PB-fg] 已导出 D:\bio_inf\LipidIN_github\PB_demo\demo\PB\pk1\figures\Pis2_quality_control.svg / .pdf / .png
```

```
[PB-fg] 图已输出到 D:\bio_inf\LipidIN_github\PB_demo\demo\PB\pk1\figures
```

图已出到 D:\bio_inf\LipidIN_github\PB_demo\demo\PB\pk1\figures (每张含 svg / pdf / png)

✓ 7 出图 完成, 耗时 4.1s

配置大模型 全流程 设计 质控 解读 假设 确认 报告 状态 帮助

输入指令, 如: 全流程 / 定位 / 4 / 状态 / 帮助

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5.5 (Optional) Configure an LLM and set the design

- 配置大模型 / `configure LLM` — a short wizard asks for provider / key / model. Needed only for natural-language design, interpretation, and hypotheses.
- 设计 / `design` , e.g. "use A1–A3 and B1–B3, A is the case group".
- 背景: `liver tumor vs adjacent` — gives the biology so hypotheses stay on-topic.
- After hypotheses appear: `确认` to start verification, or just say how to change them ("drop #2", "focus on PC").

6. What you get (outputs)

All outputs land in the `PB/pk1/` folder (and `common/pk1/` for Stage 1). The key ones:

File	Meaning
<code>*_PB_diagnostic.csv</code>	Per-peak PB diagnostic-ion hits
<code>*_PB_isomers.csv</code>	Competing double-bond isomers per species (ranked, with confidence)
<code>*_PB_dbpos.csv</code>	Finest grain: (species, f, <code>n-x</code> position) with intensity share
<code>PB_dbposition_quant.csv</code>	Quantity per (species, position) per sample (<code>__area</code> / <code>__norm</code>)
<code>PB_species_quant.csv</code>	Species-level totals (conventional layer)
<code>PB_species_diff.csv</code> / <code>PB_dbposition_diff.csv</code>	Differential analysis (species vs position layers)
<code>PB_qc.csv</code> , <code>PB_qc_replicate.csv</code>	QC + two-injection reproducibility
<code>PB_omega_index.csv</code> , <code>PB_position_profile.csv</code> , <code>PB_position_preference.csv</code>	Omega-family shares & dominant-position flips
<code>figures/Fig1..Fig2</code> (svg/pdf/png)	Publication figures
<code>agent_report/PB_agent_report.md</code>	Final Markdown report

Real demo figures produced by the `出图` step:

Fig 1 — position dimension (dominant-position flips, position-vs-amount decoupling, Δ share heatmap):

Crossing lines = the dominant double-bond position flips between groups

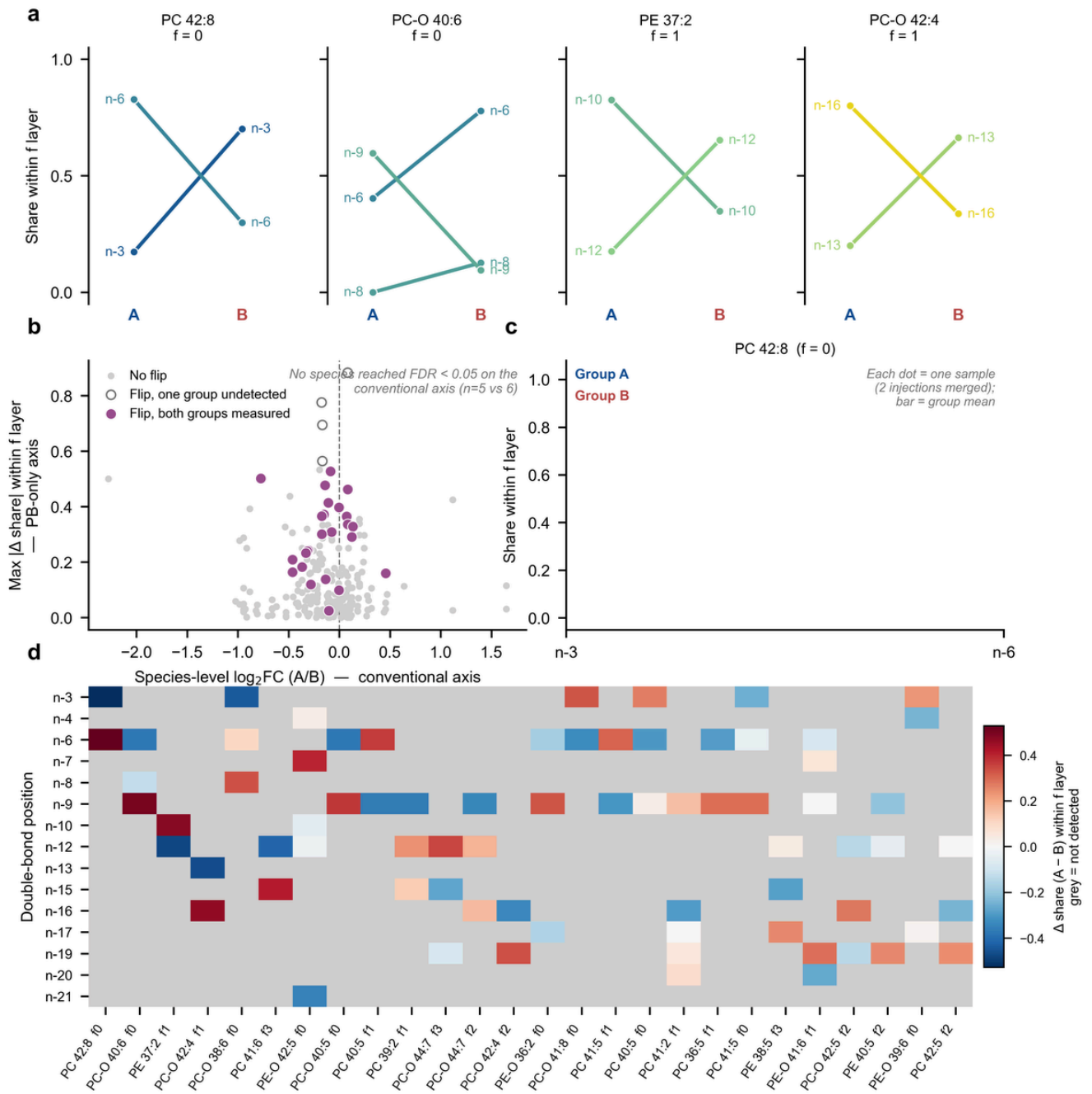
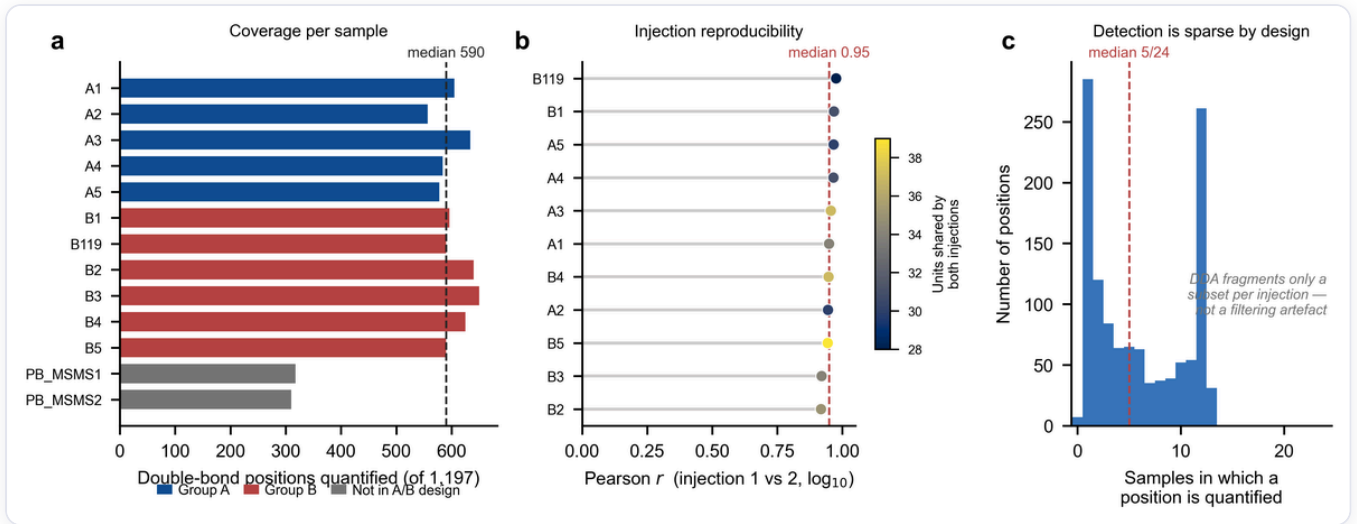


Fig 2 — quality control (coverage, injection reproducibility):



Note: In the demo above the experiment design (case/control groups) was not set, so the per-sample dot panel (Fig 1c) is intentionally blank. Run `设计` first to populate it.

7. Troubleshooting (anticipated errors)

① `ImportError: eq3_search / eq3_similarity` (C++ engine won't load).

The engines are prebuilt for **CPython 3.11, Windows x64** (`eq3_search.cp311-win_amd64.pyd` , `eq3_similarity.cp311-win_amd64.pyd`). If your interpreter is a different Python version or platform, the import fails. Fix: run with **Python 3.11 on Windows x64** (e.g. `D:\anaconda3\python.exe` if it is 3.11). `eq3_similarity` can be rebuilt from source (`build_cpp.py` / `compile_cpp_similarity.py` , needs MSVC + pybind11), but `eq3_search` has **no source in this folder** (only the `.pyd`), so it cannot be recompiled here — you must keep the shipped `.pyd` and match Python 3.11.

② **A path with a space is truncated.** When you type a folder path into the chat, the agent stops the path at the first space. `D:\... \PB library` becomes `D:\... \PB` and you get *"directory does not exist"*. Fix: put libraries/data in a **space-free path**, or set them by editing `run.py` (which handles spaces), or pass the library file paths individually where the space is inside the last segment only if unavoidable.

③ **MemoryError / machine swaps during Stage 2.** The libraries are large. Keep `EQ3_WORKERS = 1` and `raw_workers = 6` (the defaults) on 16 GB; do not raise them. Close other memory-heavy apps.

④ **"Not a DDA file" / acquisition rejected.** `tm_raw2pkl` accepts **strict DDA only**. DIA/SWATH/PASEF/AIIon files are refused by design. Re-acquire in DDA, or convert appropriately.

⑤ **"common master table not found" when starting Stage 2/3.** Stage 2 needs Stage 1's `master_table.csv` , and Stage 3 needs `master_table_quant.csv` . Run Stage 1 (`annotation`) first, with quantification enabled.

⑥ **Library file not found.** The `.pkl` libraries are **not** shipped in the repo (too large). Download them from <https://zenodo.org/records/21421866> and point the agent (`谱库目录 ...` , space-free) or `run.py` at that folder.

- ⑦ **No double-bond positions in the output.** This usually means Stage 1 (common) was skipped or the PB tolerances were changed. Keep the PB defaults (loose precursor `pb_ppm_err1=60` / `pb_da_err1=0.05` , tight fragment `pb_da_err2=0.02`) — these are deliberately set for PB diagnostic ions; tightening the precursor tolerance discards real annotations.
- ⑧ **Port already in use.** Another process holds `8790` . Start with `--port 8791` (any free port).
- ⑨ **Garbled console output / `UnicodeEncodeError`** . The entry points force UTF-8 stdout; if you invoke a module directly on a GBK console you may see mojibake — run through `run.py` / `pb_agent_server.py` , or set `PYTHONIOENCODING=utf-8` .
- ⑩ **LLM features do nothing.** Without `配置大模型` , free-form design/interpretation/hypotheses fall back to offline rules (still functional, less tailored). Configure a provider + key to enable them.
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8. Citation

LipidIN — *Nature Communications* 2025, 16:4566 (Xu, H. et al.). Spectral libraries:
<https://zenodo.org/records/21421866>